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Key Points:

- Carbon foils are a robust and proven technology for ion and neutral atom detection in space
- This paper reviews the properties of the interaction of particles with carbon and graphene foils
- Graphene foils may soon replace carbon foils and open new opportunities for space research

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Carbon foils for space plasma instrumentation

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Abstract Carbon foils have been successfully used for several decades in space plasma instruments to detect ions and neutral atoms. These instruments take advantage of two properties of the particle-foil interaction: charge conversion of neutral atoms and/or secondary electron emission. This interaction also creates several adverse effects for the projectile exiting the foil, such as angular scattering and energy straggling, which usually act to reduce the sensitivity and overall performance of an instrument. The magnitude of these effects mainly varies with the incident angle, energy, and mass of the incoming projectile and the foil thickness. In this paper, we describe these effects and the properties of the interaction. We also summarize results from recent studies with graphene foils, which can be made thinner than carbon foils due to their superior strength. Graphene foils may soon replace carbon foils in space plasma instruments and open new opportunities for space research in the future.

1. Introduction

Thin carbon foils have become a critical component for many space plasma instruments. They are used for coincidence and time-of-flight (TOF) measurements in plasma spectrometers [e.g., McComas *et al.*, 2007], plasma mass spectrometers [e.g., Wüest, 1998], and energetic neutral atom (ENA) imagers [e.g., McComas *et al.*, 1991; Funsten *et al.*, 2009; Fuselier *et al.*, 2009; Gruntman, 1997; Wurz, 2000; Krimigis *et al.*, 2004; Mitchell *et al.*, 2000; Pollock *et al.*, 2000]. The interaction of ions and neutral atoms, which we hereafter refer to as “particles,” with the foil results in two processes that can be exploited for particle detection and measurement of their properties: (1) particle-induced secondary electron emission, which enables coincidence and time-of-flight measurements, and (2) particle charge state modification, which enables detection and measurement of energetic neutral atoms (ENAs).

The interaction of the primary particle with the carbon foil also creates several adverse effects that can impact the detection and measurement of the particles. These effects include the following: (1) angular scattering, which perturbs the particle trajectory; (2) energy straggling, which reduces the particle energy; and (3) modification of particle charge state into one that detracts from (rather than enables) the measurement. A standard approach to minimize these effects is to use the thinnest foils possible. The current state of the art is carbon foils with a nominal areal density of $\sim 0.5 \mu\text{g cm}^{-2}$.

Carbon foils are made from pure carbon that is arc evaporated onto glass slides that are coated with a surfactant. Because the carbon deposition rate is monitored using a quartz crystal microbalance, the nominal carbon foil thickness is typically quoted as an areal density ($\mu\text{g cm}^{-2}$). Thicknesses from 0.5 to 3 $\mu\text{g cm}^{-2}$ are typical for space plasma instruments. Assuming a foil composition of only carbon and a density of about 1.8 to 2 g cm^{-3} [Ritzau and Baragiola, 1998; Stoner, 1991], this corresponds to a thickness range from roughly ~ 2.5 to 17 nm.

These foils can span several cm^2 and thus are sufficiently thin to require a support grid, which is traditionally electroformed nickel. The carbon foil is transferred from the slide to the support grid by floatation on water (sometimes mixed with alcohol to adjust the surface tension) and picked up by the grid holder with the support grid. Once dried, its performance can be characterized, and it can be installed in an instrument. The Arizona Carbon Foil Company Inc. is the standard supplier for commercial carbon foils thinner than $\sim 2.0 \mu\text{g cm}^{-2}$.

Figure 1 shows scanning electron microscope (SEM) images of a nominal $0.5 \mu\text{g cm}^{-2}$ carbon foil supported by a 333 lines per inch (lpi) nickel grid. On the left-hand side, the carbon foil in three cells appears ripped in a manner that indicates that the foil is mounted to the grid through a process that introduces an intrinsic tension. A

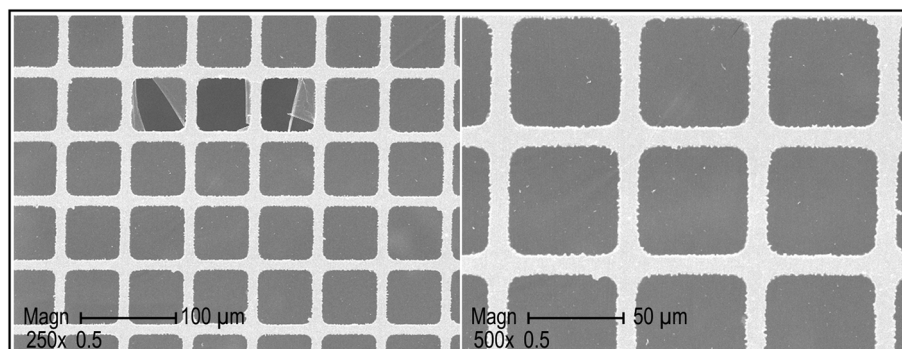


Figure 1. Scanning electron microscope images of a nominal $0.5 \mu\text{g}/\text{cm}^2$ carbon foil at two different scales as indicated by the bar.

defect, such as a slight tear, propagates throughout the grid cell and causes retraction of the carbon foil toward the cell edges. The nickel support grid acts as a reinforcement that limits tearing to individual cells; this greatly enhances foil durability and survival during launch (see *McComas et al.* [2004] and section 1.1).

A nominal thickness is the foil thickness quoted by the manufacturer. Carbon foils are neither fabricated nor typically used in ultrahigh vacuum conditions. Therefore, foils likely contain bulk adsorbates as well as surface adsorbates. *Funsten and Shappiro* [1997] suggested that an initially high sputter rate of foils is due to adsorbates, and *Allegrini et al.* [2006, 2015b] showed that the adsorbate layers of impurities increase the nominal thickness by ~ 1 to $1.5 \mu\text{g}/\text{cm}^2$. Adsorbates are typically governed by the partial pressure of ambient volatiles; this is likely the largest uncertainty between instrument calibration in the laboratory and foil performance in space. For simplicity and clarity, we use the nominal value of the carbon foil thickness throughout this paper.

Recently, a new type of thin, freestanding carbon foil has emerged—graphene. Graphene is a two-dimensional (2-D) carbon honeycomb lattice. Graphene being the strongest material on Earth [*Lee et al.*, 2008], its thickness can be controlled to a single atomic layer and, thus, tremendously advance foil performance by reducing angular scattering and energy straggling. Recent studies [*Allegrini et al.*, 2014, 2015a; *Ebert et al.*, 2014] have demonstrated superior performance of graphene and thus their potential utility for space instrumentation. We will summarize some of these findings in this paper in section 4.

The interaction of particles with carbon foils is a broad topic with extensive literature, for example, for use as targets in accelerators. We focus here on the aspects of the particle-foil interaction that are relevant to space plasma measurements and measurement performance. The knowledge of this interaction is directly applicable to performance modeling for development of new instrumentation concepts, instrument design, and as a critical component of end-to-end instrument performance models that are used for data analysis and interpretation. The energy range of interest ranges from the minimum energy required for a particle to pass through a foil (several hundred eV) up to hundreds of keV.

Foils similar to those that are the focus of this study have been used in accelerators (e.g., diamond-like carbon, or DLC [*Lichtenstein et al.*, 1997]) and to attenuate ultraviolet light or low-energy particles in space (e.g., aluminum-carbon-polymer or silicon-polyimide-carbon composite [*Mitchell et al.*, 2000]). DLC has also been used in space in a time-of-flight mass analyzer [*Jvkova et al.*, 1995]. While this paper does not cover composite or DLC foils, the principles and validation techniques used to understand the amorphous carbon foils that are the focus of this paper can be directly applied to these specialized foils.

The structure of this paper is as follows. In section 1.1, we present a summary of the flight experience with carbon foils. In section 2, we describe the relevant properties of the interaction of ions/ENAs/UV radiation with carbon foils. We discuss how these properties may depend on each other in section 3. We summarize recent findings from the interaction of ions with graphene foils in section 4. Our conclusions and future outlook are stated in section 5.

1.1. Carbon Foils: A Proven Technology

Carbon foils, despite being extremely thin, are remarkably resistant to the stresses that they are subject to during a rocket launch. They have been successfully operating in space instruments for decades.

Table 1. Summary of Flight Experience With Carbon Foils (as of 1 February 2016)

Mission/Instrument	Launch Date	Number of Carbon Foils	Size (mm)	Thickness ($\mu\text{g cm}^{-2}$ (or nm))	Operation Time (Through 1 February 2016)
Ulysses/SWICS	10/6/1990	1	4 × 80	~3	~18.7 years
CRRES/LOMICS ^d	7/25/1990	3	—	0.5	0.5 years
Geotail/STICS	7/24/1992	3	—	~2	23.5 years
Wind/SWICS ^a	11/1/1994	1	—	~2	5.6 years
Wind/STICS	11/1/1994	3	—	~2	21.3 years
Wind/MASS	11/1/1994	1	4 × 15	2	21.3 years
Interball/Tail SKA-1	8/3/1995	1	—	~10 nm	~10 h
SOHO/MTOF	12/2/1995	1	4 × 15	2.2	~20.7 years
Equator-S/ESIC	12/2/1997	16	~10 × 29	3.3	~0.4 years
ACE/SWICS	8/25/1997	1	4 × 80	1.5	~18.5 years
ACE/SWIMS	8/25/1997	1	4 × 15	<2	~6.3 years
Cassini/CAPS/IMS ^b	10/15/1997	8	7 × 18	1.1	13.8 years
DS-1/PEPE	10/24/1998	8	—	~5 nm	3 years
IMAGE/MENA	3/25/2000	15	10 × 16	0.6–0.7	5.8 years
IMAGE/LENA	3/25/2000	—	—	~2	5.8 years
Cluster2/CODIF	7/16/2001 and 8/9/2001	4 × 16	~10 × 29	3.4–3.5	~17 years × 3
Messenger EPPS/FIPS	8/3/2004	1	Ø 40	1.0	~7.9 years
New Horizons/SWAP	01/19/2006	1	Ø 16	1.0	~10 years
STEREO A and STEREO B ^c /PLASTIC	10/26/2006	2 × 1	—	3.5	A: 9.3 years; B: 7.9 years
TWINS A and B	06/28/2006 and 3/13/2008	2 × 10	10 × 16	0.5	A: 9.6 years; B: 7.9 years
IBEX-Hi	10/19/2008	15 conversion foils; two detector foils	~21 × 49; Ø 56	0.6; 2	7.3 years
IBEX-Lo	10/19/2008	16	—	2.5	7.3 years
Juno/JADE-I	8/5/2011	12	~14.4 × 9.3	0.6	~0.2 years
Van Allen Probes/HOPE	8/30/2012	5	~8 × 20	~0.6	3.4 years
MAVEN/STATIC	11/18/2013	48	—	<1.0	1.4 years
MMS/HPCA	3/13/2015	4 × 16	~8 × 10	~1	0.9 years

^aWind/SWICS was turned off on 27 May 2000 due to a high-voltage failure.

^bCassini/CAPS was turned off in June 2011 due to an electric short. It was powered on again in March 2012 and remained on for 78 days before another electrical short occurred, shutting down the instrument again.

^cSTEREO B lost contact on 1 October 2014. Contact has not been reestablished as of 31 August 2015.

^dCRRES was lost on 12 October 1991.

Table 1 summarizes the flight experience with carbon foils. Such a table was originally published in *McComas et al.* [2004]. Here we update operating durations of those instruments and add new instruments from recent missions.

From our experience, the yield (fraction of foils suitable for flight) is <50% and depends on the carbon foil thickness and size, quality of the procured carbon foils, quality of the support grid, consistency of the foil floating process, and the carefulness of subsequent handling and storage of the mounted foils. As an example, we have found higher yield of foils when the support grid is selected from the center of ~20 cm square stock grid.

Screening of carbon foils for flight instruments typically includes a bulk measurement of particle transmission. We expect that most foil defects occur as pinholes or tears that span one or more adjacent cells, as in Figure 1 (left). Because the foils are extremely thin and highly transparent, the use of optical light transmission is challenging for foil characterization. Instead, a broad, low-energy heavy ion beam (e.g., 5 keV Ar⁺) that illuminates a foil is highly scattered in regions in which foil is present and is unscattered in regions with pinholes or tears, thus providing high contrast and a quantitative measure of pinhole fraction [Funsten et al., 1992a, 1992b]. Foils with >99% carbon coverage can be routinely achieved [e.g., Funsten et al., 2009].

Carbon foil screening also typically includes foil thickness measurements. Again, optical methods are challenging because the foils are too thin. However, measurement of the angular scatter distribution, and specifically the angular half width, of a narrow H⁺ beam is a quantitative proxy for thickness that additionally provides direct information on foil performance [Funsten et al., 1992b, 1993a]. Details on the angular scatter distribution of projectiles are provided in section 2.3.

While carbon foils are typically <100 atomic layers thick, they have proven to be tremendously robust. Recent measurements of the resistance to static pressure showed that carbon foils of 1 $\mu\text{g}/\text{cm}^2$ can withstand static

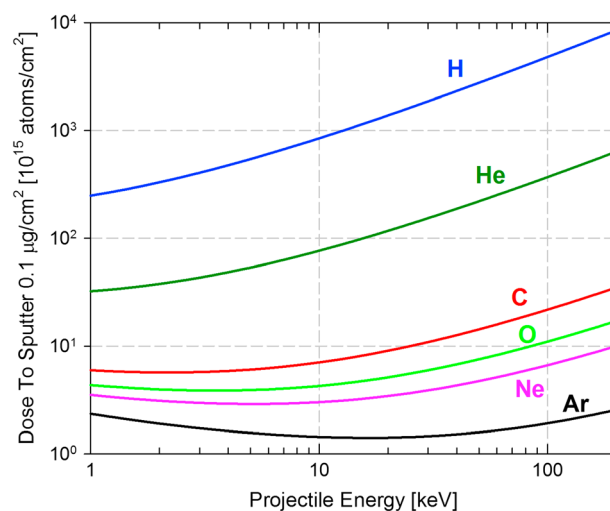


Figure 2. Foil erosion rates due to sputtering by H, He, C, O, Ne, and Ar projectiles. These results are scaled to the Ar + sputter measurements of *Funsten and Shappirio* [1997] by the incident nuclear stopping power of these projectiles in carbon. Doses for significant foil sputtering are enormous in comparison to total dose of particles entering a plasma spectrometer or ENA imager over an entire mission. As an example, using solar wind at 1 keV/amu it takes ~7800 years to remove 0.1 $\mu\text{g}/\text{cm}^2$ ($<1/10$ th of the foil thickness) at a constant flux of 10^6 ions/ cm^2 s on the foil.

pressures up to 9 kPa (1.3 psi) with almost no damage when supported by 13.1 lines per mm (333 lines per inch) grids [Allegri and Dayeh, 2014]. The carbon foils act as a hermetic seal, and thus, proper venting of the instrument is typically required to ensure that the foils survive the rapid change of pressure during rocket ascension. The measurements also showed that the amount of damage to a foil with improper venting would be very quickly limited to a small fraction ($<10\%$) of the entire foil because the freshly created holes in the foils act as a vent, and the foils are no longer a hermetic seal.

Over the lifetime of a mission, foils can be exposed to significant fluxes of incident particles, which can sputter both the entrance and exit foil surfaces. Sputtering is generally associated with Coulombic energy loss processes and thus has a linear dependence on the nuclear stopping power S_N [Sigmund, 1969]. *Funsten and Shappirio* [1997] measured the sputtering of carbon foils due to incident Ar^+ beams at 20 keV and 40 keV, which have similar

nuclear stopping power at these incident energies. Their results showed a carbon foil sputter rate of $0.7 \mu\text{g}/\text{cm}^2$ at a dose of $10^{15} \text{ Ar}^+/\text{cm}^2$, which corresponds to a sputter yield of 3.5 carbon atoms per incident Ar^+ assuming the foil is entirely carbon. Using the dependence of sputter yield on stopping power S_N and values of S_N derived from SRIM [Ziegler et al., 2010], we scale the Ar results to incident H, He, C, O, and Ne projectiles. For each one of these projectiles, Figure 2 shows the dose required to remove $0.1 \mu\text{g}/\text{cm}^2$ of foil, at which point we expect that the foil performance may change. H and He, which are the most dominant species in space plasmas, have the lowest sputter rate and require an enormous dose for any significant sputtering. For all possible scenarios involving projectiles at energies > 1 keV, we expect that sputtering is negligible. One possible scenario for which little data exists and risk to foils is potentially high lies in exposure of a foil to high density of thermal oxygen ions, and thus chemical etching, in the ram direction of a spacecraft at altitudes in the vicinity of 400–800 km. This requires further research.

2. Properties of the Interaction of Particles With Carbon Foils

2.1. Electron Emission

The interaction of particles with a carbon foil leads to secondary electron emission (SEE) from both sides of the foil [e.g., Baragiola, 1994; Rothard et al., 1990b; Hasselkamp, 1992]. As schematically illustrated in Figure 3, the incident particle transfers energy to the electrons of the solid through excitation and ionization processes [Sternglass, 1957]. The energized electrons that are close to a foil surface and with outward trajectories with respect to the surface may have enough energy to overcome the foil's work function and escape from the foil. These secondary electrons, which can be easily detected, provide information that an incident particle passed through the foil.

Of particular importance for the use of foils for coincidence and time-of-flight measurements is the property of the secondary electron emission (SEE), in particular its average yield, yield distribution, angular distribution, energy distribution, energy-angle dependence, and dependence on incident atomic species. These are addressed in the following sections. Omitted from this review are the UV-induced photoelectron emission from carbon foils that can be a background in instruments with foils that are exposed to space [Hsieh et al., 1980] and electron-induced secondary electron emission from carbon foils [e.g., Hölzl and Jacobi, 1969].

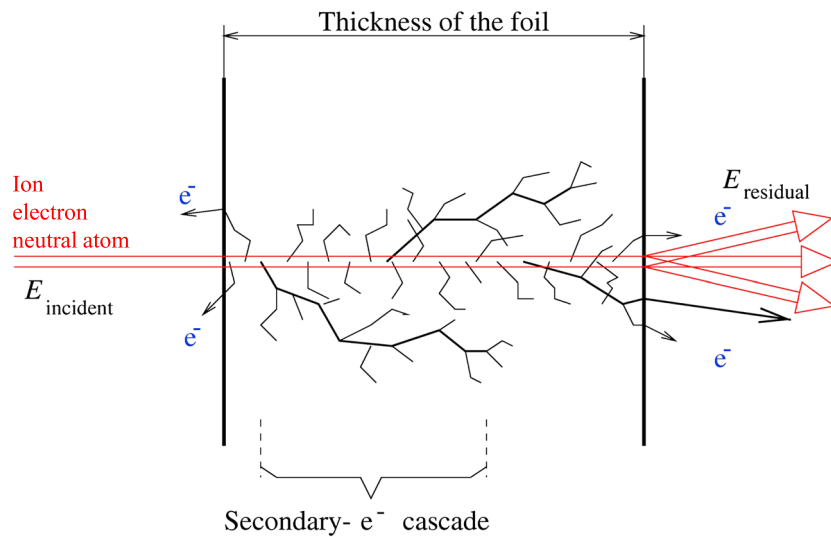


Figure 3. Schematic of the secondary electron emission process.

2.1.1. Secondary Electron Yield

The secondary electron yield (SEY), γ , is the average number of secondary electrons emitted per incident particle. The SEY from carbon foils depends mainly on the primary particle type, its energy, its angle of incidence with respect to the surface, and the surface cleanliness and roughness. To a lesser extent, other parameters such as temperature and bulk foil composition may influence SEE. The SEY of atomic particles incident on a foil target is generally found to be proportional to the electronic stopping power S_e (originally formulated by Sternglass [1957] for particles incident on solid targets). For example, Rothard *et al.* [1990b] showed a remarkable proportionality (within a factor of 2) of the total secondary electron yield from carbon foils to the electronic stopping power over a wide range of energies (15 keV/amu to 46 MeV/amu) and species ($1 \leq Z_1 \leq 92$).

The electronic stopping power is the energy loss (due to distant collisions with the electrons of the target) per unit length of target material. It is therefore independent of the foil thickness. The proportionality between the stopping power and the SEY implies that the energy deposited contributes to the average number of electrons. However, when the thickness of the foil becomes comparable to the mean-free path of the electrons, Sternglass's theory predicts a decrease in the yield [Sternglass [1957]; Frischkorn *et al.* [1983]; Ritter [1985]]. This has been observed experimentally at energies of the order of ~ 1 MeV and for foils rather on the thick side for space plasma instrumentation [Garnir *et al.* [1981]; Rothard *et al.* [1995]; Frischkorn *et al.* [1983]]. Nevertheless, carbon foils down to $0.5 \mu\text{g}/\text{cm}^2$, where this decrease should be observed, still produce enough secondary electrons [e.g., Ritzau and Baragiola, 1998] for the detection of particles, as supported by the extensive flight experience shown in Table 1. While the yields are obviously sufficient for ultrathin foils, this is an area of the literature where additional measurements are needed.

When a particle traverses the foil, both the entrance and exit surfaces of the foil emit secondary electrons: the yield from the entrance surface, γ_b , is called the backward yield, whereas the yield from the exit surface, γ_f , is called the forward yield. The ratio $R = \gamma_f/\gamma_b$ is usually slightly above 1 at energies > 1 keV/amu [e.g., Ritzau and Baragiola, 1998] due to momentum conservation arguments (more secondary electrons with momentum in the same direction of the incident particle). However, R can drop significantly below unity depending on the carbon foil thickness, particle energy, and particle species [see Allegrini *et al.*, 2003]. The observed drop in R is attributed to the energy loss of the particle between the entrance and exit surfaces and a resulting reduced S_e at the exit surface relative to the entrance surface. For an incident particle of mass m in amu, atomic number Z , and energy E , S_e increases with increasing E to a maximum value at particle speed $Z^{2/3}v_B$ [Lindhard and Scharff, 1961] where v_B is the Bohr velocity, which corresponds to an energy $mZ^{4/3} \cdot 25.3$ keV. Particles below this energy can experience substantial energy loss in the foil and thus a substantially reduced S_e and γ_f relative to γ_b . However, this effect is significant only when the energy loss in the foil is substantial.

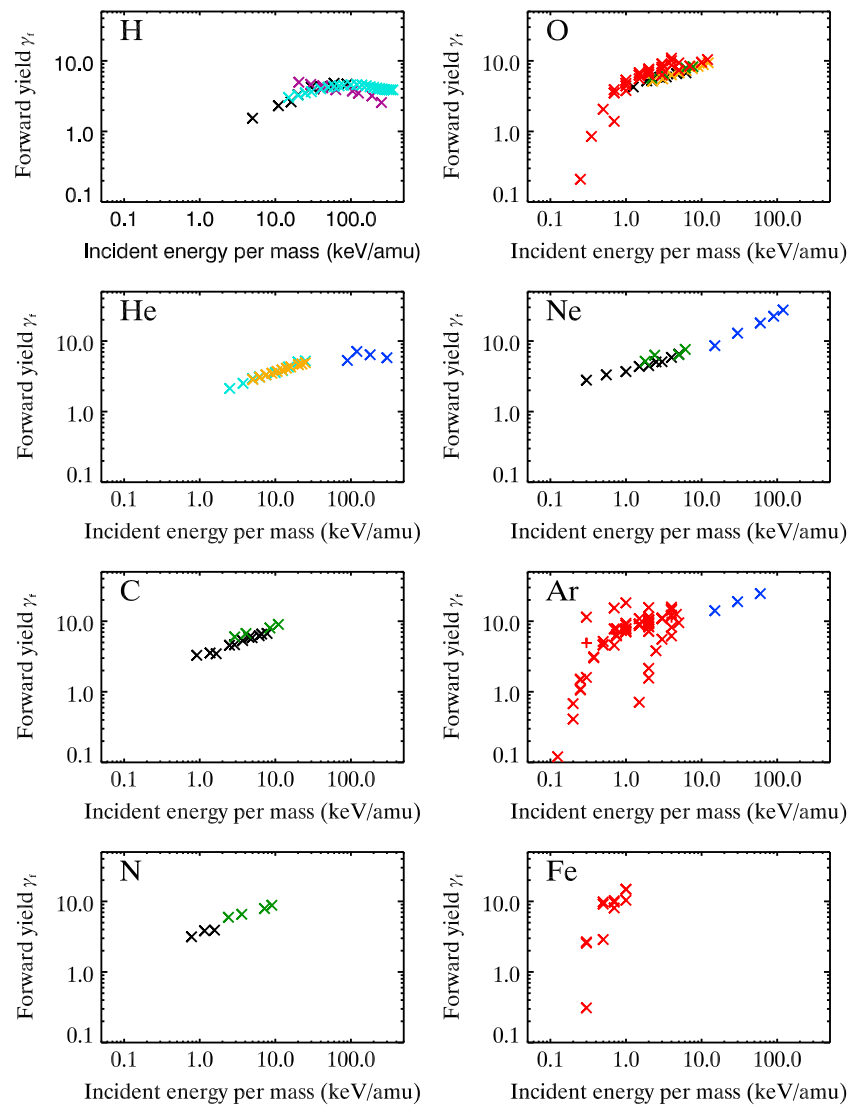


Figure 4. Forward electron yields from carbon foils for different species from Ritzau and Baragiola [1998] (black), Keller et al. [1991] (green), Allegrini et al. [2003] (red), Rothard et al. [1990b] (blue), Kerkow et al. [1999] (orange), Gelfort et al. [1997] (turquoise), and Meckbach et al. [1975] (purple).

Because the SEY is proportional to S_e , we expect a strong dependence of SEY on the angle of incidence θ of a particle relative to the surface normal direction. Numerous studies show that the SEY increases with a $1/\cos(\theta)$ dependence on the angle of incidence [e.g., Ferrón et al., 1981]. For projectiles that scatter significantly in the foil, we expect a strong angular dependence of the SEY at the exit surface, although this is complicated by effects of energy loss in the foil.

The literature on average electron yield from ion beams incident on solid targets is abundant. However, the studies specifically related to carbon foils at keV energies are comparatively sparse. There are a number of measurements taken with different carbon foils (thickness, manufacturer, etc.), projectile species, and energies. Ritzau and Baragiola [1998] measured yields for the first 10 atomic species from a nominal $0.5 \mu\text{g}/\text{cm}^2$ carbon foil. Allegrini et al. [2003] reported on SEYs from iron projectiles on carbon foils of different thicknesses. We show these yields and those from other sources in Figures 4 and 5 for the forward and backward yields, respectively, as a function of energy per unit mass. There is general agreement within about a factor of 2 between these yields, suggesting that projectile speed is a general ordering parameter for SEY. A practical tool to estimate the yields would be to use the same expressions that fit the electronic stopping power data and adjust the parameters to fit the secondary electron yields data. It may be possible to estimate the yields

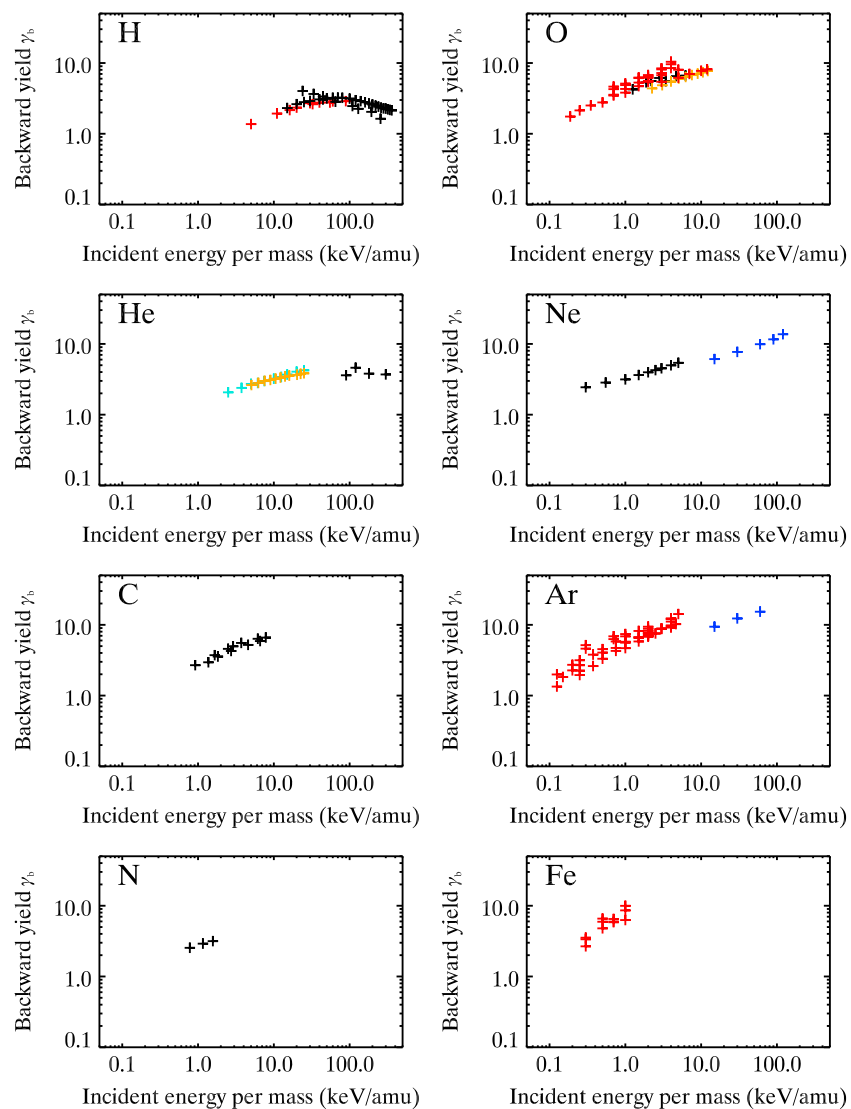


Figure 5. Backward electron yields from carbon foils for different species from Ritzau and Baragiola [1998] (black), Allegrini et al. [2003] (red), Rothard et al. [1990b] (blue), Kerkow et al. [1999] (orange), and Gelfort et al. [1997] (turquoise).

from carbon foils for projectiles, for which there is no data available, by using yields from solids where the literature is more abundant.

The secondary electron emission yield depends strongly on the cleanliness of the surface. A freshly sputter-cleaned surface generally has a lower electron yield. Carbon foils retain impurities, which has been mostly attributed to water adsorbed on the surface. For example, Allegrini et al. [2003] showed that the yield decreases with particle fluence. Even under high vacuum ($P < 7 \cdot 10^{-8}$ mbar) over a few hours without beam bombardment, the adsorbates build up and the yields go up again, though not as high as right after pump down or when the foil is bombarded for the first time. Similar reports also confirm the decrease in yield as a function of fluence [e.g., Rothard et al., 1996; Svensson and Holmén, 1981].

In plasma instruments operated in space, even though the pressures on the outside of the spacecraft can become lower than typical vacuum in testing chambers on Earth, components inside the instrument continue to outgas. Because sputtering by incident particles is negligible, adsorbate layers, which govern the foil work function, are present at abundances that are in equilibrium with the partial pressure of outgassing volatile species. The Rosetta Orbiter Spectrometer for Ion and Neutral Analysis mass spectrometer on the Rosetta mission acquired background spectra throughout the mission [Schläppi et al., 2010], indicating water

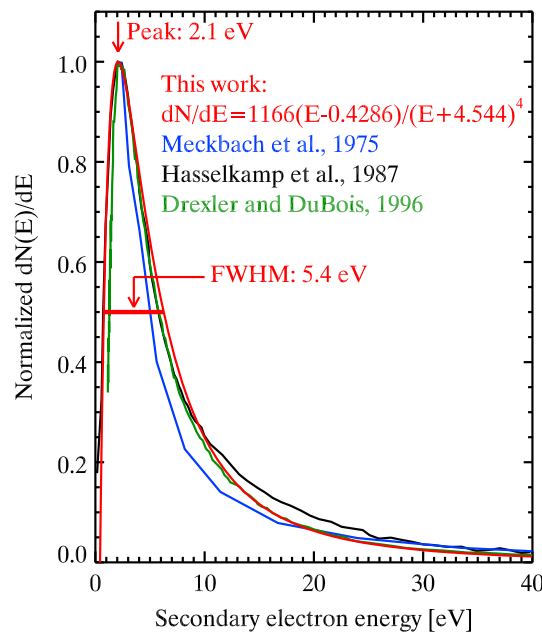


Figure 6. Secondary electron energy distribution from the proposed expression in red compared to measurements from Meckbach *et al.* [1975] in blue, Hasselkamp *et al.* [1987] in black, and Drexler and DuBois [1996] in green.

the Fermi energy of the solid, and Φ is the work function. Chung [1975] later refined this expression to include quantum mechanical reflection at the solid-vacuum interface.

Without specific knowledge of E_F and Φ , we adapt an empirical expression based on the formula from Chung and Everhart:

$$\frac{dN}{dE} = k \frac{(E - a_0)}{(E - a_1)^4}$$

where E is the electron energy in eV, k is a normalization constant (e.g., $k = (a_0 - a_1)^{\frac{3}{4}}$ for a maximum of 1 of the distribution), $a_0 = 0.4286$, and $a_1 = -4.544$. This distribution has a peak at 2.1 eV and a FWHM of 5.4 eV.

Figure 6 shows this distribution and compares it to other studies. The measurements by Meckbach *et al.* [1975] were obtained from 41.5 keV protons hitting a 5 or 10 $\mu\text{g}/\text{cm}^2$ carbon foil. Drexler and DuBois [1996] measured the secondary electron energy from the forward and backward emission at different angles for 1 MeV protons passing through a 10 $\mu\text{g}/\text{cm}^2$ carbon foil. Here we report their result of the backward emission for an unspattered (i.e., uncleaned) foil at 45° with respect to the foil normal. Our approximation above provides a reasonable estimate of the energy distribution.

2.1.3. Emission Angle-Energy Dependence

The angular distribution of secondary electrons follows a cosine dependence with respect to the surface normal [e.g., Jahrreiss and Oppel, 1972]. However, the energy and emission angle distributions of secondary electrons are correlated. Measurements by Drexler and DuBois [1996] clearly show differences in the peak and width of the energy distributions for different angles of the secondary electron emission distribution. The secondary electrons were emitted from sputter-cleaned 5 $\mu\text{g}/\text{cm}^2$ carbon foil from incident 1.5 MeV protons. A somewhat similar observation was made by Pferdekämper and Clerc [1975] with a 2 $\mu\text{g}/\text{cm}^2$ carbon foil. While both of these reports were obtained with faster particles than the energy range of interest for this review, we expect similarity to SEE for lower energy particles because the process of secondary electron formation in the solid, and its escape from the solid are similar. For the sake of modeling secondary electrons in a space plasma instrument, it would be useful to collect the relevant data of the energy-angle dependence for carbon foils. Meanwhile, one could use the theories on electron emission [e.g., Schou, 1980, 1988] to couple the energy and angle distributions.

is the predominant outgassing species from spacecraft outgassing as well as hydrocarbons, nitrogen-containing compounds, and other minor species. Lack of knowledge about the adsorbate composition on foil surfaces of instruments operated in space is likely the largest uncertainty of SEE and thus of the absolute sensitivity of the instrument.

2.1.2. Energy Distribution

The energy distribution of ion-induced secondary electrons is strongly peaked between 1 and 5 eV with a long tail at higher energies. Rothard *et al.* [1990a] measured the energy distribution for secondary electron emission from 1.2 MeV protons incident on a sputter-cleaned 1500 Å carbon foil and found a maximum yield at 2.1 ± 0.3 eV. The SEE peak contains ~85% of all emitted electrons. Hasselkamp *et al.* [1987] also reported a peak SEE energy at 2 eV from a solid carbon surface and a full width at half maximum of 5.4 eV.

Chung and Everhart [1974] proposed a simple analytic formula for the energy distribution of secondary electrons: $dN/dE = (E - E_F - \Phi)/(E - E_F)^4$

where E is the secondary electron energy, E_F is

2.1.4. Emission Statistics

The secondary electron emission statistics play a role in the probability of detection of the secondary electrons by a detector and as a consequence in the sensitivity of an instrument [e.g., *Hovestadt et al.*, 1995; *Funsten et al.*, 2009]. The probability, η , of triggering a signal on the MCP is a strong function of the number of emitted electrons when the expectation value of this number is low. For the sake of illustrating this with an example, we assume a Poisson distribution for the number of emitted electrons. One can show that the probability to trigger a signal on the detector is given by $\eta = 1 - \exp(-p \cdot \gamma)$, where p is the detection efficiency (e.g., ~ 0.6 for microchannel plates) and γ is the average secondary electron yield. The probability η decreases very quickly for low values of γ . Therefore, the knowledge of the emission statistics for low yields can be critical for a correct calculation of the detection efficiencies.

The assumption of a Poisson distribution is not correct as has been shown experimentally [*Lakits and Winter*, 1990; *Gruntman et al.*, 1990; *Kozochkina et al.*, 1993]. The distribution of the number of secondary electrons is broader than Poisson, and the probability of emitting no electrons is larger than the value $P_0 = e^{-\gamma}$ that would result from a Poisson distribution. Other distributions, such as a Polya distribution [*Dietz and Sheffield*, 1975], a sum of Poisson distributions [*Vicanek and Urbassek*, 1993], or even a Gaussian distribution [*Kurz et al.*, 1992] have been proposed.

2.2. Charge State Modification

As an incident ion or atom traverses a foil, it undergoes multiple charge exchange events. For the energies and foil thicknesses of interest for plasma spectroscopy, the exit charge state distribution has reached an equilibrium and retains no memory or imprint of its incident charge state. ENA imaging is critically dependent on conversion of ENAs to ions for their subsequent electrostatic energy analysis and detection [e.g., *McComas et al.*, 1991; *Funsten et al.*, 2009]. Additionally, the performance of foil-based TOF mass spectrometers can be degraded if an electric field is present in the TOF drift region, in which ions exiting in different charge states follow different trajectories and thus different times of flight.

Exit charge state distributions from foils have been extensively studied [e.g., *Betz*, 1972; *Shima et al.*, 1992]. The exit charge state distribution depends on foil composition [*Phillips*, 1955]. Because our focus is on carbon foils for space plasma spectroscopy and ENA imaging, we do not consider here results obtained for other foil materials or the extensive studies published for the charge state distribution of MeV incident ions incident on foils.

At low energies, the charge exchange processes in the bulk foil and at its exit surface become adiabatic, and exiting atoms are mostly or wholly neutral. As the projectile energy increases, the equilibrium charge state distribution becomes positive as electrons are stripped off of the projectile. At MeV energies and higher, the particles can be fully stripped.

Ions with strong electron affinity are observed to exit a carbon foil with significant fraction of negative ions [*Kallenbach et al.*, 1995; *Funsten*, 1995]. Even helium has been observed to exit as a metastable negative ion [*Funsten et al.*, 2001]; while the negative ion fraction is small, the lifetimes of the metastable states of He^- are sufficient for these species to remain negative ions over timescales of a typical TOF measurement. Trajectories of negative ions in plasma spectrometers should be considered from conceptual design of an instrument to its use in space.

Table 2 shows a comprehensive summary of measurements of charge state distributions in the general energy range applicable to plasma spectroscopy and ENA imaging. Many of these references delve deeply into theoretical explanation and empirical representation of the observed charged state distributions. Notably, the ions and energies of interest in space physics, including planetary space environments, are largely covered. These data are essential for understanding performance of foil-based space instrumentation and development of new instrument concepts.

Two gaps in the table are notable: low-energy N charge state distributions and Kr charge state distributions. Figure 7 shows the charge state distributions for N and Kr for nominal $0.5 \mu\text{g cm}^{-2}$ carbon foils obtained using the identical experimental configuration of *Funsten et al.* [1993a]. No negative ions were observed for either N or Kr, and no Kr^{2+} ions were observed up to 60 keV.

2.3. Angular Scattering

Angular scattering of a keV particle in a foil degrades performance in several ways, the most important of which are extreme particle scattering, so they do not strike a detector and introduction of uncertainty in

Table 2. Summary of Charge Distribution Measurements Applicable to Plasma Spectrometry and ENA Imaging

Species	Z	Mass (amu)	Energy Range (keV)		Reference
			Minimum	Maximum	
H	1	1	2	50	<i>Allegri et al.</i> [2014]
			0.3	6	<i>Funsten et al.</i> [2009]
			0.5	20	<i>Kallenbach et al.</i> [1993]
			1	30	<i>Funsten et al.</i> [1993b]
			20	230	<i>Kreussler and Sizmann</i> [1982]
			60	300	<i>Meggitt et al.</i> [1973]
D	1	2	64	313	<i>Dettmann et al.</i> [1974]
			8	100	<i>Berkner et al.</i> [1972]
He	2	4	10	20	<i>Allegri et al.</i> [2014]
			1	30	<i>Funsten et al.</i> [1993a]
			6	30	<i>Bürge et al.</i> [1990]
			64	313	<i>Dettmann et al.</i> [1974]
B	5	11	3	22	<i>Kallenbach et al.</i> [1995]
			6	22	<i>Bürge et al.</i> [1993]
			266	600	<i>Lennard et al.</i> [1981]
C	6	12	10	50	<i>Allegri et al.</i> [2014]
			2	16	<i>Kallenbach et al.</i> [1995]
			6	18	<i>Bürge et al.</i> [1993]
			9	24	<i>Bürge et al.</i> [1990]
			88	380	<i>Christensen et al.</i> [1978]
N	7	14	198	267	<i>Lennard et al.</i> [1981]
			369	1450	<i>Smith and Whaling</i> [1969]
			5	10	<i>Allegri et al.</i> [2014]
			21	35	<i>Bürge et al.</i> [1990]
			89	479	<i>Hvelplund et al.</i> [1970]
O	8	16	232	338	<i>Lennard et al.</i> [1981]
			269	313	<i>Dettmann et al.</i> [1974]
			372	1450	<i>Smith and Whaling</i> [1969]
			10	50	<i>Allegri et al.</i> [2014]
			1	30	<i>Funsten et al.</i> [1993a]
F	9	19	5	80	<i>Kallenbach et al.</i> [1995]
			10	32	<i>Bürge et al.</i> [1990]
			89	479	<i>Hvelplund et al.</i> [1970]
			267	420	<i>Lennard et al.</i> [1981]
			4	57	<i>Kallenbach et al.</i> [1995]
Ne	10	20	303	490	<i>Lennard et al.</i> [1981]
			10	29	<i>Bürge et al.</i> [1993]
			91	482	<i>Hvelplund et al.</i> [1970]
			10	40	<i>Bürge et al.</i> [1990]
			80	482	<i>Hvelplund et al.</i> [1970]
Na	11	23	307	571	<i>Lennard et al.</i> [1981]
			11	105	<i>Bürge et al.</i> [1993]
			16.5	44	<i>Bürge et al.</i> [1990]
			17	69	<i>Bürge et al.</i> [1993]
			90	482	<i>Hvelplund et al.</i> [1970]
Mg	12	24	394	527	<i>Lennard et al.</i> [1981]
			7	72	<i>Gonin et al.</i> [1995]
			80	465	<i>Hvelplund et al.</i> [1970]
Al	13	27	411	626	<i>Lennard et al.</i> [1981]
			8	135	<i>Kallenbach et al.</i> [1995]
			90	481	<i>Hvelplund et al.</i> [1970]
Si	14	28	463	760	<i>Lennard et al.</i> [1981]
			10	98	<i>Kallenbach et al.</i> [1995]
			14	84	<i>Bürge et al.</i> [1993]
P	15	31	476	646	<i>Lennard et al.</i> [1981]
			529	854	<i>Lennard et al.</i> [1981]
S	16	32	85	425	<i>Christensen et al.</i> [1978]
			10	96	<i>Kallenbach et al.</i> [1995]
			16	96	<i>Bürge et al.</i> [1993]
			544	757	<i>Lennard et al.</i> [1981]

Table 2. (continued)

Species	Z	Mass (amu)	Energy Range (keV)		Reference
			Minimum	Maximum	
Cl	17	35	7	70	<i>Kallenbach et al.</i> [1995]
			18	70	<i>Bürgi et al.</i> [1993]
			70.2	330.5	<i>Turkenburg et al.</i> [1975]
			581	812	<i>Lennard et al.</i> [1981]
Ar	18	37	74		<i>Bürgi et al.</i> [1993]
		40	10	50	<i>Allegrini et al.</i> [2014]
			15	100	<i>Bürgi et al.</i> [1990]
			20	100	<i>Bürgi et al.</i> [1993]
			56	450	<i>Hvelplund et al.</i> [1970]
			70.7	325.4	<i>Turkenburg et al.</i> [1975]
			180	1450	<i>Smith and Whaling</i> [1969]
			685	1099	<i>Lennard et al.</i> [1981]
K	19	39	20	78	<i>Bürgi et al.</i> [1993]
			666	1000	<i>Lennard et al.</i> [1981]
Ca	20	40	8	200	<i>Gonin et al.</i> [1995]
			685	953	<i>Lennard et al.</i> [1981]
Sc	21	45	783	1207	<i>Lennard et al.</i> [1981]
Ti	22	48	10	144	<i>Gonin et al.</i> [1995]
V	23	51	896	1207	<i>Lennard et al.</i> [1981]
Cr	24	52	16	208	<i>Gonin et al.</i> [1995]
Cr	24	52	896	1437	<i>Lennard et al.</i> [1981]
Mn	25	55	949	1411	<i>Lennard et al.</i> [1981]
Fe	26	56	28	168	<i>Bürgi et al.</i> [1993]
			190	1450	<i>Smith and Whaling</i> [1969]
			987	1206	<i>Lennard et al.</i> [1981]
Ni	28	59	18	236	<i>Gonin et al.</i> [1995]

the distance of travel of a particle in a time-of-flight mass spectrometer. Angular scattering of atomic projectiles results from their Coulombic interactions with nuclei of atoms of the foil. The angular scattering of ions through carbon foils has been investigated for many ions (H, D, He, Li, C, N, O, Ne, Ar, K, Ca, Cr, and Fe) and foil thicknesses from 1.5, 2.0, 2.5, 3.5, 4.5, 5.7, 7.8, and 15.6 $\mu\text{g}/\text{cm}^2$ [Høgborg *et al.*, 1970; Bernstein *et al.*, 1970; Lüthi, 2003] at a few keV up to 100 keV. The only measurements for foils of nominal thickness down to 0.5 $\mu\text{g}/\text{cm}^2$ were published by Funsten *et al.* [1993a] for H, He, and O ions, and Ebert *et al.* [2014] for H, He, C, N, O, Ne, and Ar.

One of the results of a semi-empirical theory describing small-angle multiple scattering derived by Meyer [1971] is that the angular scattering half width at half maximum, $\psi_{1/2}$, is proportional to the incident energy, E_0 , for a given foil and projectile. This has been verified experimentally [e.g., Høgborg *et al.*, 1970; Bernstein *et al.*, 1970; Funsten *et al.*, 1993a; Lüthi, 2003; Ebert *et al.*, 2014] and is illustrated in Figure 8. Funsten *et al.* [1993a] defined the foil constant, k_F , as a key performance parameter for predicting angular scattering over a large energy range:

$$k_F = \psi_{1/2} E_0$$

k_F is unique for a particular projectile-foil combination and can be used as a proxy to measure foil thickness (typically with H^+ as the incident projectile) as well as to interpolate and extrapolate performance over a broad range of projectile species and energies. Table 3 gives foil constants for H, He, C, N, O, Ne, and Ar for the carbon foil in Ebert *et al.* [2014].

Allegrini *et al.* [2016] recently showed that the foil constant of 0.5 $\mu\text{g}/\text{cm}^2$ foils can be reduced by a factor of ~ 1.2 by annealing the foils in an argon atmosphere at 500 $^\circ\text{C}$ for an hour.

2.4. Energy Loss and Straggling

In the laboratory frame, incident projectiles transfer kinetic energy to a foil through both nuclear (Coulombic) and electronic processes. The electronic stopping dominates at higher energies ($> \sim 1$ keV/amu) and reaches a maximum at an energy E_{MAX} , which corresponds to the Bragg peak as well as a maximum in secondary

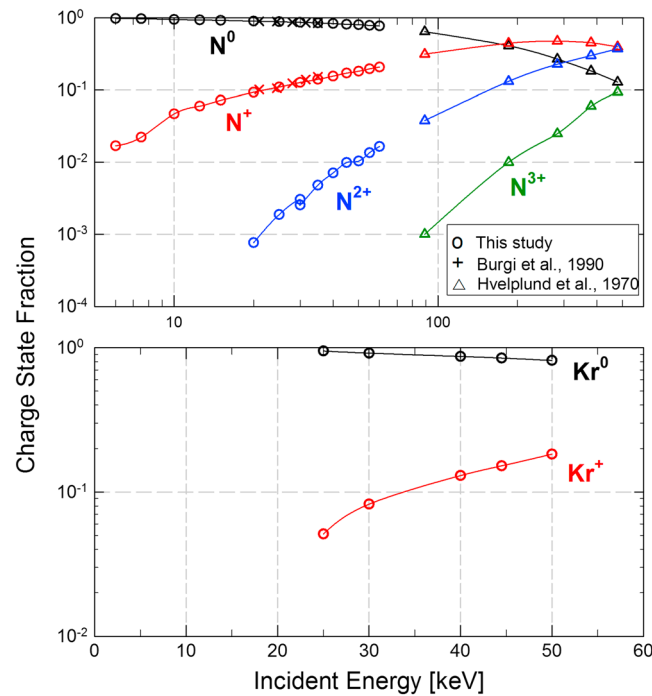


Figure 7. Exit charge state distributions for N and Kr for nominal $0.5 \mu\text{g cm}^{-2}$ carbon foils.

energy. Thus, the energy loss is a distributed value around a most probable energy loss. The energy loss of ions through carbon foils has been measured for different foils thicknesses (0.5, 1.0, 1.5, 2.0, and $3.0 \mu\text{g/cm}^2$), incident ions (H, He, C, N, O, Ne, and Ar), and incident ion energies (1 to 50 keV) for incident ion trajectories along the foil normal [Allegrini et al., 2006, 2015b]. Other sporadic measurements at these energies and for similar foil thicknesses can also be found but are generally for different foil materials (Al, Au, and Si) and rarely for carbon.

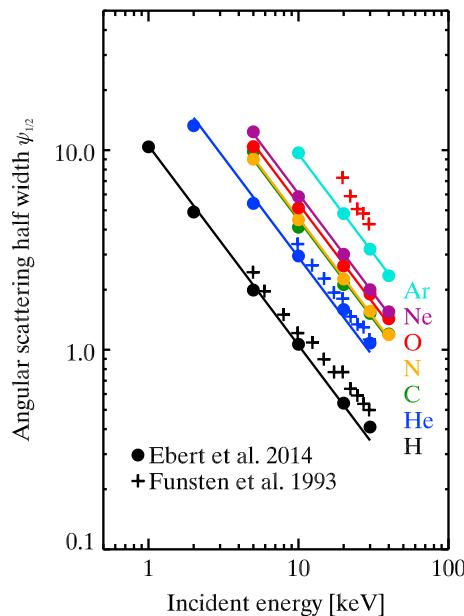


Figure 8. Scattering half width of H, He, C, N, O, Ne, and Ar as a function of incident energy for nominal $0.5 \mu\text{g/cm}^2$ carbon foils [from Funsten et al., 1993a; Ebert et al., 2014]. The foil constants for the different species for the carbon foil from Ebert et al. [2014] are given in Table 3.

electron yield [Rothard et al., 1990b]. Lindhard and Scharff [1961] provide an approximation $E_{\text{MAX}} \sim mZ^{4/3} * 25.3 \text{ keV}$ where m and Z are the particle mass (in amu) and atomic number, respectively, although, in comparison to SRIM tables [Ziegler et al., 2010], this approximation underestimates E_{MAX} by a factor of three for H, He, and Li and overestimates E_{MAX} for species heavier than Ne. The nuclear stopping only contributes when the particles are almost stopped in the target material. Below the E_{MAX} , the electronic stopping power, which is the energy lost to electronic processes by the projectile per unit path length into the material, behaves like a frictional force and increases linearly with increasing projectile velocity. Above the E_{MAX} , the electronic stopping power decreases with increasing energy as the interaction time with individual electrons in the foil decreases.

Since collisions are a statistical process, not all ions lose the same amount of energy. Thus, the energy loss is a distributed value around a most probable energy loss. The energy loss of ions through carbon foils has been measured for different foils thicknesses (0.5, 1.0, 1.5, 2.0, and $3.0 \mu\text{g/cm}^2$), incident ions (H, He, C, N, O, Ne, and Ar), and incident ion energies (1 to 50 keV) for incident ion trajectories along the foil normal [Allegrini et al., 2006, 2015b]. Other sporadic measurements at these energies and for similar foil thicknesses can also be found but are generally for different foil materials (Al, Au, and Si) and rarely for carbon.

Allegrini et al. [2015b] derived a straightforward empirical method to estimate the energy loss and straggling of hydrogen by measuring the scattering half width and applying semiempirical relationships. The areal mass density Nd (in $\mu\text{g/cm}^2$), which is also an estimate of the foil thickness, is given by

$$Nd = 0.153 \cdot k_F,$$

where N is the atomic density of the foil material, d is the thickness of the foil, and k_F is the foil constant in keV degree as derived from its scattering distribution. The energy loss then becomes

$$E_0 - E_1 = k \frac{Nd}{a_0 + a_1 E_0^{-0.4} + a_2 E_0^{0.25}},$$

where E_0 and E_1 are the incident and residual energy in keV, respectively, $k = 19.9$ (per atom of ^{12}C) is a unit

Table 3. Scattering Constants for Different Species for the 0.5 $\mu\text{g}/\text{cm}^2$ Carbon Foil in Ebert *et al.* [2014]

Species	H	He	C	N	O	Ne	Ar
kF (keV $^\circ$)	11.1 ± 1.5	29.1 ± 2.7	44.8 ± 3.5	45.8 ± 1.2	58.8 ± 2.8	60.4 ± 1.4	95.7 ± 1.3

conversion factor, $a_0 = -37.19$, $a_1 = 178.1$, and $a_2 = 9.236$. The straggling, Ω , which is the full width at half maximum of the energy distribution after the carbon foil (assuming a monoenergetic incident beam) is given by a similar expression:

$$\Omega = b_0(kNd)^{b_1} E_0^{b_2},$$

where $b_0 = 8.3 \cdot 10^{-3}$, $b_1 = 0.70$, and $b_2 = 0.54$. Similar relationships for heavier ions are in preparation.

2.5. UV Attenuation and Photoemission Yield

Carbon foils attenuate UV light to which particle detectors are sensitive. Hsieh *et al.* [1980] measured the attenuation of the Lyman alpha (121.6 nm) and He I (58.4 nm) resonance lines by carbon foils of different thicknesses. They fit their data and data from other sources with power laws: transmittance is $0.26 \exp(-0.37 \cdot x)$ for Ly α and $0.1 \exp(-0.56 \cdot x)$ for He I, where x is the thickness in $\mu\text{g}/\text{cm}^2$. They also report the forward secondary electron yield for the Ly α : $2.6 \cdot 10^{-3} \exp(-0.38 \cdot x)$, which is much less than that from ions (>1) at keV energies.

2.6. Molecular Ions Traversing Carbon Foils

Generally, molecular ions passing through carbon foils break apart and exit the foil as individual atomic fragments. However, under special conditions (molecules of light atoms, in thin foils at sufficiently high energy), a small fraction (e.g., ~ 0.1 to $\sim 10\%$) of incident molecular species can exit the foil intact as molecular ions. Escovitz *et al.* [1979] measured ~ 1 – 10% of H_2^+ after 15 to 40 $\text{\AA}$ carbon foils at 12.5 keV/amu for an H_2^+ beam. Funsten *et al.* [1994] measured the charge state fractions and angular scattering of exiting products from H^+ , H_2^+ , and H_3^+ beams interacting with thin carbon foils (nominal 0.3 to 2 $\mu\text{g}/\text{cm}^2$). Exiting H_3^+ or H_3^- were not observed. Interestingly, they found that the fraction of H_2^+ exiting the foil is inversely proportional to the square of the scattering half width of atomic H: the fraction of H_2^+ ranges from 1.5% at 1.5° to less than 0.1% at 7° . This inverse relationship is evidence for reformation of H_2 at the exit surface by recombination of H fragments in close proximity.

Atomic fragments of incident molecules generally follow the same scatter distribution as an incident atomic projectile at the incident molecular speed. This indicates that atomic fragments from their molecular parent interact independently with the foil. However, the energy loss of molecular ions and their fragments is for the most part not equal to the sum of the energy loss of the individual fragments at the incident molecular speed. Ray *et al.* [1992] measured the energy loss of H^{n+} clusters passing through 3, 6, and 10.5 $\mu\text{g}/\text{cm}^2$ carbon foils at energies above that of the electronic stopping power maximum (E_{MAX}). They found that the total energy loss is less (more) than that for the corresponding isolated protons below (above) ~ 50 keV/proton. Similarly, Levi-Setti *et al.* [1982] measured that the energy loss per H fragment of H^{2+} (12.5 and 25 keV/amu) is consistently lower (by 10–20%) than that of protons of the same velocity passing through 0.3 to 1.1 $\mu\text{g}/\text{cm}^2$ carbon foils. Also, the total energy loss approaches that of two independent protons as the foil thickness decreases.

Similarly, the secondary electron emission of molecular ions is different than the sum of the emission from the constituents at the same velocity. Hasselkamp and Scharmann [1983] measured the electron yields from a solid carbon surface induced by a range of ions and molecules between about 40 keV and 1 MeV. While the yield for H_n^+ is lower than n times that of H^+ at the lower velocities of their experiment, it is opposite at the higher velocities. Moreover, they also found that the electron yields of the molecular ions N_2^+ and CO^+ are always less than the sum of the yields of the independent constituents at equal velocities in the energy range studied. A study by Billebaud *et al.* [1996] with H_n^+ clusters between 40 and 120 keV/proton also shows that the electron yield from carbon foils is lower for the clusters than it is for the sum of the individual constituents at the same velocity. Lakits *et al.* [1989] measured the electron yields, the electron energy distribution, and the electron emission statistics of clean polycrystalline gold bombarded with H^+ , H_2^+ , and H_3^+ ions between 1 and 5 keV/amu. The yields for the molecular ions are also less than that of the sum from the single

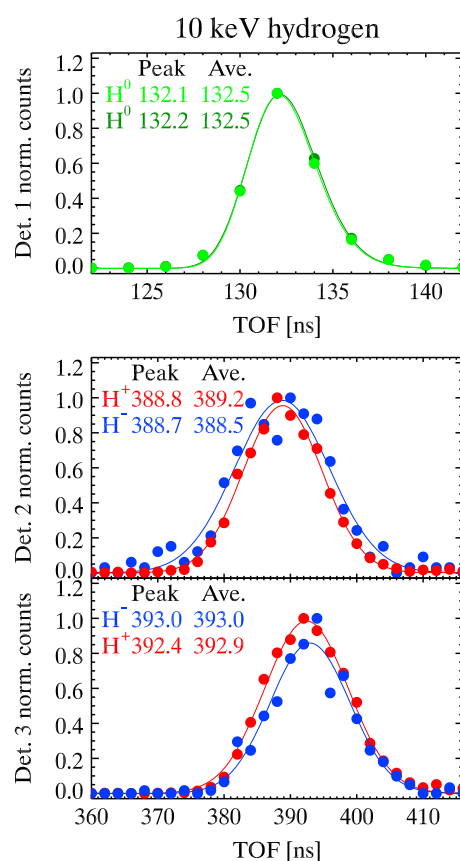


Figure 9. Raw TOF spectra (dots) fit with Gaussian curves of 10 keV hydrogen after passing through a $1.0 \mu\text{g}/\text{cm}^2$ carbon foil. (top) The neutrals. (middle and bottom) The H ions (H^+ or H^-) following different paths depending on their charge state.

constituents. Finally, *Ritzau and Baragiola* [1998] measured similar ratios for electron yields from $0.5 \mu\text{g}/\text{cm}^2$ for incident molecules (H_2 , OH, H_2O , CO, N_2 , and O_2). The effect is observed for both the backward and the forward emission, although it is weaker for the forward emission.

As can be seen from the few examples above, the interaction of hydrogen molecular ions with carbon foils is complex. In general, the topic of the interaction of molecular ions with carbon foils is understudied. This topic would benefit from more research on targeted molecular species (e.g., molecular ions specific to a science question) as well as systematic characterizations of the different aspects of the interaction with many different species.

3. Modeling of Carbon Foil Effects in Space Plasma Instrument and Interdependence of the Properties of the Interaction

Modeling the interaction of particles with carbon foils in a space plasma instrument requires taking into account most or all of the processes described in section 2. While some processes enable a measurement, e.g., secondary electron emission for TOF mass spectrometry and ENA ionization by charge modification, other process degrade the measurement, e.g., energy loss and angular scattering. Each of these processes should be considered in an instrument's conceptual design; models of the processes should be included as integral components of an instrument's

end-to-end performance model; and these models should be validated through flight calibration and flight data to ensure quantitative understanding of performance over the mission lifetime.

In practice, the processes described in section 2 are generally described using probabilistic models and can have substantial uncertainty and significant inter-dependence. For example, adsorbates on the foil surfaces changes throughout a mission as instrument and spacecraft outgassing decreases; changes in adsorbate composition and abundance between calibration and flight, which we do not currently understand, can modify the work function of the foil and thus secondary electron emission properties as well as the exit charge state distribution. Interdependency of processes and quantitative evaluation of their uncertainties is a rich target for future study for better quantification of instrument performance.

The energy loss of particles in foils depends on the exit angle of the ion, which can be a strong function of angular scattering. *Famá et al.* [2000a, 2000b, 2002] measured and modeled the energy loss of protons and deuterium as a function of exit angle for Al, Au, and Si foils. The model shows that the increase in energy loss at larger angles is not only due to a longer travel distance in the foil (cosine effect) but also due to the roughness of the foil surface (which was found to be the dominant term for exit angles up to $\sim 25^\circ$) and due to the additional energy loss from the Coulombic collisions with foil atoms that causes large angle of scattering.

The angular scattering distribution seems to have no dependence on the exit charge state as shown by *Funsten et al.* [1993a] with H^0 and H^+ exiting a $0.5 \mu\text{g}/\text{cm}^2$ carbon foil, at least within the resolution of the measurement.

Similarly, we find no correlation between the exit charge state distribution and energy loss in the foil. Figure 9 shows raw TOF histograms (dots) for H^- , H^0 , and H^+ exiting a carbon foil from a 10 keV hydrogen beam at normal incidence on a $1.0 \mu\text{g}/\text{cm}^2$ carbon foil. Also shown are simple Gaussian fits to the TOF spectra.

Figure 9 (top) shows the TOF of neutral H (with nominal drift distance 206.3 mm), Figures 9 (middle) and 9 (bottom) show the TOF of H^+ and H^- ions (with nominal drift distances 545.7 and 550.0 mm, respectively). Since there is no systematic difference in Figures 9 (middle) and 9 (bottom), we conclude that there is no dependence of the energy loss on the exit charge state within the resolution of the experiment. This is reasonable, because most theories of charge modification of ions transiting foils attribute the final charge state distribution to electronic processes at or near the exit surface. Therefore, this interaction can only be at most a small fraction of the total energy lost during transit through the foil.

As already mentioned in section 2.1, the angular distribution of secondary electrons is correlated with their energy distribution. The roughness of the surface is known to decrease the secondary electron yield, but it may also affect the angular and energy distributions.

The relative importance of each of these effects needs to be evaluated on a case by case basis. Some designs are more sensitive to a particular process than others. For example, all aspects of the secondary electron emission in the IBEX-Hi detector section [Funsten *et al.*, 2009; Allegrini *et al.*, 2008] are critical parameters because electrons are electrostatically steered toward a CEM detector from the interior surfaces of a large cylinder, whereas energy straggling and charge exchange in the detector section is less important because it is a coincidence measurement rather than a TOF measurement. Conversely, the angular scattering and the charge exchange properties are very important at the entrance subsystem where incident ENAs are converted to ions for their subsequent electrostatic energy measurement and coincidence detection.

TRIM (Transport of Ions in Matter) [Ziegler *et al.*, 2010] provides a robust tool for estimating energy loss and angular scattering in foils, as well as estimating the electronic stopping power at the entrance and exit surfaces (for prediction of secondary electron emission). However, the largest uncertainty in TRIM calculations and simulations is our lack of quantitative knowledge of foil composition and thickness. Carbon foils are mostly not only made of carbon but also contain other elements, such as, e.g., H, O, F, Na, and K [Gruntman, 1997 and references therein; Allegrini *et al.*, 2016]. These impurities can originate during foil fabrication, as residue from the parting agent from the transfer process from slide to support grid and from adsorbates. Assuming that H (from water and hydrocarbons) and O (from water and oxidized impurities) are the dominant elemental impurities, we constructed foils in TRIM with different compositions: atomic abundances of C (88%) and O (12%) [Allegrini *et al.*, 2016] and different atomic abundances of C and H_2O . We then used TRIM to attempt to reproduce angular scattering and energy loss data in a consistent fashion and compared the simulation results with laboratory measurements. Our findings are the following: (1) It is possible to determine a thickness in TRIM for which the simulated energy loss would match the measurements of 1–50 keV H passing through a nominal $0.5 \mu\text{g}/\text{cm}^2$ carbon foil, but only over a limited energy range; (2) TRIM usually overestimated the energy straggling compared to the measurements; and (3) the foil thickness in TRIM that would match the H angular scattering measurements would not accurately predict angular scattering for other species (He, C, N, O, Ne, and Ar).

Thus, when available, empirical data of foil performance or subsystem performance should be used instead of TRIM. When empirical data are not available, TRIM remains the most useful and practical tool for predicting foil performance.

End-to-end instrument performance models can be exceedingly complex when they rely on empirical or theoretical models of multiple foil processes. Furthermore, the complexity of quantitative measurements, which can (a) target a wide range of particle species, energies, and angles of incidence on a foil and (b) include complex electron optics within the measurement subsystems, likely results in sparse quantitative knowledge of performance. One method to quantify and monitor the performance of plasma instruments that use coincidence or TOF measurements is to report count rates of the start and stop detectors (including noncoincident events) along with the coincident count rates [Funsten *et al.*, 2005]. These can be used to track the absolute detection efficiency of the coincidence or TOF subsystem throughout a mission. As an example, the HOPE mass spectrometers [Funsten *et al.*, 2013] on the Van Allen Probes provide continuous monitoring of the absolute detection efficiency of the TOF subsystem. This includes processes such as the emission, electrostatic collection, and detection of secondary electrons emitted by incident particles; effects of angular scattering and energy loss of incident particles; and effects of the exit charge state distribution on particle trajectories in the drift region, which is not field free. Data such as this can be used to validate models of processes described in section 2 and understand their impact on instrument measurements.

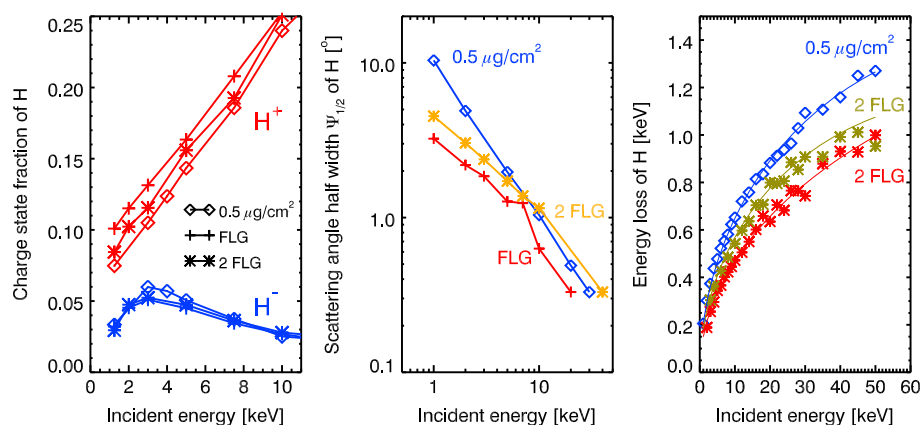


Figure 10. Comparisons between few layer graphene (FLG), 2xFLG, and carbon foils ($0.5 \mu\text{g}/\text{cm}^2$): (left) charge state fraction of hydrogen after the foil [Allegrini et al., 2014], (middle) scattering angle half width of hydrogen [Ebert et al., 2014], and (right) average energy loss of hydrogen after passing through the foil [Allegrini et al., 2015a].

4. Graphene: A New Type of Foil With “Improved” Performance

From a space science perspective, graphene’s thin nature and considerable strength (Young modulus $\sim 1 \text{ TPa}$) and elasticity (spring constant $\sim 1\text{--}5 \text{ N m}^{-2}$) make it a highly attractive candidate foil material that may further minimize some of the undesirable effects associated with thin carbon foils. Any improvement compared to thin carbon foils could provide at least two benefits for particle instruments in space. Instruments could be made smaller without loss of sensitivity, and more sensitive instruments are possible with equivalent resources.

Initial work has shown that (see Figure 10) (1) the charge state distributions of 1–50 keV ions (H, He, C, N, O, and Ar) exiting graphene foils and carbon foils are similar [Allegrini et al., 2014], (2) the angular scattering of ions (H, He, C, N, O, Ne, and Ar) exiting graphene foils is reduced by up to a factor of 3 when compared to state-of-the-art ($0.5 \mu\text{g}/\text{cm}^2$) carbon foils [Ebert et al., 2014], and (3) the average energy loss of ions (H, He, C, N, and O) is less for graphene foils than for thin carbon foils, while the energy straggling is larger [Allegrini et al., 2015a].

The increased energy straggling from graphene foils has been attributed to the thickness nonuniformity [Allegrini et al., 2015a]. Similarly, the apparent slower decrease of the angular scattering with energy (Figure 10, middle) for graphene is also a consequence of the thickness nonuniformity and an artifact of the experiment [Allegrini et al., 2015a]. The production and the transfer process of the graphene at the time of those studies created large differences in the thickness of the graphene foil. The coverage was also much lower than what is desirable. However, since these studies, progress has been made in the production and transfer of the graphene. We anticipate that large format graphene foils with superior performance to carbon foils will soon be available.

While there may be some concerns that graphene foils made too thin would not produce enough secondary electrons, the fact that the projectile loses energy via collisions implies that some electron emission will take place. Moreover, the fact that graphene foils in Figure 10 show a $\sim 20\%$ decrease in energy loss with respect to carbon foils does not imply that the stopping power of graphene is lower than that of amorphous carbon. Since the electron yield is proportional to the electronic stopping power, there is no reason to assume that graphene foils could generate lower yields except for the fact that Sternglass’s theory predicts a decrease of the yield when the foil thickness becomes comparable to the mean-free path of the electrons. This should be a topic of research. If the yields were lower for graphene, then the important question would be how much decrease an instrument can tolerate until the benefits of decreased thickness (e.g., scattering) are balanced by the decrease in sensitivity resulting from lower electron yields.

The initial results support further investigation of graphene as a foil candidate for space-based plasma instruments. Here we give three examples of the potential impact of substituting thin carbon foils with graphene on the design of foil-based plasma instruments: (1) Reduced angular scattering results in higher sensitivity at low energies because more particles reach the detector; (2) reduced energy loss enables lower postacceleration

and thus lower bias applied to a TOF drift region, which means lower risk, less resources, smaller high-voltage gaps (smaller instruments), etc; and (3) reduced energy loss and angular scattering also reduces the energy threshold that can be measured by an instrument; for example, it may be possible to combine two ENA instruments that are currently required to cover a broad energy range into a single instrument.

5. Conclusions

The use of carbon foils in space plasma mass spectrometry and for ENA detection and measurement now spans several decades. Not only is the technique robust but also it is at the heart of many key science advancements.

Despite this huge success, there are still gaps in our understanding of particle-foil interaction physics. Some aspects of this interaction, and the interdependence between them, are still poorly understood. Since some aspects directly impact the quality of space data analysis, a poor modeling or interpretation of this interaction can lead to erroneous interpretation of data acquired from the instrument. It is, therefore, necessary to keep expanding the body of knowledge and to continue exploring the more complex aspects of the interaction physics.

Understanding of the interaction of particles with carbon foils can enable new measurement concepts and enables higher accuracy and precision of the measurements of plasma instruments. While the particle-foil system has been extensively studied, there remain critical gaps, especially in the areas of correlations between these processes as well as their uncertainties. Even incremental advance can enable new measurements or fewer resources. An energetic neutral atom (ENA) or time-of-flight (TOF)-based instrument requiring less resources and cost for similar performance will be beneficial to any future heliospheric and planetary science missions.

Graphene foils may soon replace carbon foils in these instruments. There are still a number of challenges to overcome and tests to perform before it is qualified for flight. It is, however, a very promising material that should open new opportunities for space research in the future.

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